1 Question (15 points)

Write a recursive method

```
int aAfterB(String s)
```

which returns the number of 'a' characters succeeding a 'b' character in the string s.

2 Question (20 points)

- Write the function **dice**

  `int dice()`

  which randomly throws a 24 sided-dice and returns the result.

- Write the function **pair**

  `boolean pair(int dice1, int dice2)`

  which returns true if the values of the two dices have no divisors in common.

- Write the function **gcd**

  `int gcd(int dice1, int dice2)`

  which returns greatest common divisor of two dices.

- Write the function **diceGame**

  `void diceGame()`

  which plays the following dice game using the above functions. There are two players. The first player throws two 24-sided dices. The second player throws two 24-sided dices. Then the dices of players are compared according to this rule: If one of the player has a pair and the other not, the one with pair wins. If both have pairs, no player wins. If none of the players has a pair, the one with the largest gcd wins.

```
First player: 12 18
Second player: 5 7
Second player wins

First player: 12 18
Second player: 2 16
First player wins

First player: 5 18
Second player: 7 16
No one wins
```

3 Question (15 points)

Write a recursive method

```
int numberOfOnes(int N)
```

which returns number of 1's in N's binary representation.

```
numberOfOnes(16)
Returns 1: (10000) = 16
numberOfOnes(63)
Returns 6: (111111) = 63
```

4 Question (20 points)

Write a Java method **sortAlternate**, which takes a 2d array A as argument and a file name, then sorts the rows of the arrays alternately. The method will print the matrix into the file. First half of the array is sorted in increasing order, second half in decreasing order. Write helper methods

sortIncreasingOrder(int[][] A, int rowNo)

sortDecreasingOrder(int[][] A, int rowNo),

which sorts the row of the 2d-array A with index *rowNo* in increasing and decreasing order respectively. Then use those helper methods in **sortAlternate**.

$$A = \begin{array}{|c|c|c|c|}\hline 3 & 7 & 12 & 1 \\ \hline 1 & 8 & 4 & 6 \\ \hline 6 & 14 & 0 & 2 \\ \hline 8 & 2 & 7 & 12 \\ \hline \end{array}$$

$$A' = \begin{array}{|c|c|c|c|}\hline 1 & 3 & 7 & 12 \\ \hline 1 & 4 & 6 & 8 \\ \hline 14 & 6 & 2 & 0 \\ \hline 12 & 8 & 7 & 2 \\ \hline \end{array}$$

5 Question (15 points)

Write a method **paste**, which takes two arrays A and B and two values k, m as arguments, and returns a new array which inserts the array B at positions k and m in the array A and returns the resulting array.

```
A =  {2, 4, 5, 7, 8}, B = {8, 10, 12}
k = 1, m = 3.

{2, 8, 10, 12, 4, 5, 8, 10, 12, 7, 8}

A =  {3, 4, 8, 9}, B = {7, 10, 12, 20, 24}
 k = 0, m = 3.

{7, 10, 12, 20, 24, 3, 4, 8, 7, 10, 12, 20, 24, 9}
```

6 Question (15 points)

Suppose that we are modeling AmiralBattı game. Write the Java method **int numberOfNonHits(boolean[][] shots, boolean[][] shipParts)**, which counts the number of ship parts that are not hit. In this function; shots[i][j] is true, if we have shot the cell in the i'th row and j'th column; false otherwise. Yet in this function; shipParts[i][j] is true, if there is a ship part in the cell in the i'th row and j'th column.

shots =

true	true	false	false
false	false	false	false
true	true	false	false
false	false	false	false

shipParts =

true	false	true	true
false	false	false	false
false	true	true	true
false	false	false	false

numberOfNonHits returns 4.